

4.10.16

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Question

The area of the region bounded by the curve $y = x + 1$ and the lines $x = 2$ and $x = 3$ is:

- ① $\frac{7}{2} sq. units$
- ② $\frac{9}{2} sq. units$
- ③ $\frac{11}{2} sq. units$
- ④ $\frac{13}{2} sq. units$

Solution

$y = x + 1$ can be expressed as

$$\mathbf{n}^T \mathbf{x} = c \quad (1)$$

$$\mathbf{n}^T = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \&c = -1 \quad (2)$$

where $\mathbf{n}^T$ is the normal vector of the line.

Let the points of intersection of the curve $y = x + 1$ and the lines $x = 2$ and $x = 3$ be **A** and **B**.

Solution

For **A**:

$$\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad (3)$$

The augmented matrix of the above is:

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ -1 & 1 & 1 \end{array} \right). \quad (4)$$

Then,

$$\mathbf{R}_1 \longrightarrow \mathbf{R}_2 + \mathbf{R}_1 \quad (5)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \end{array} \right) \quad (6)$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}. \quad (7)$$

Solution

For **B**:

$$\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad (8)$$

The augmented matrix of the above is:

$$\left(\begin{array}{cc|c} 1 & 0 & 3 \\ -1 & 1 & 1 \end{array} \right). \quad (9)$$

Then,

$$R_2 \longrightarrow R_2 + R_1 \quad (10)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 4 \end{array} \right) \quad (11)$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}. \quad (12)$$

Solution

Hence the points of the quadrilateral are $(2, 0)$, $(2, 3)$, $(3, 4)$ and $(3, 0)$ (which are **A**, **B**, **C** and **D** respectively.)

Area of the quadrilateral can be calculated by using the formula,

$$Ar = \frac{1}{2} \|(\mathbf{A} - \mathbf{C}) \times (\mathbf{D} - \mathbf{B})\| \quad (13)$$

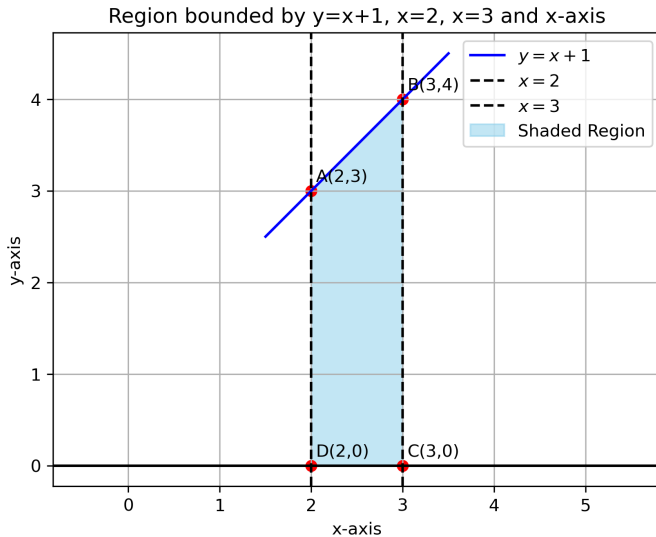
$$(\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -1 \\ -4 \end{pmatrix} \quad (14)$$

$$(\mathbf{B} - \mathbf{D}) = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad (15)$$

$$(\mathbf{A} - \mathbf{C}) \times (\mathbf{D} - \mathbf{B}) = \begin{vmatrix} -1 & 1 \\ -4 & 3 \end{vmatrix} = -3 - 4 = -7 \quad (16)$$

$$\therefore Ar = \frac{1}{2} |-7| = \frac{7}{2}. \quad (17)$$

Therefore (option 1) $\frac{7}{2} sq. units$ is the correct answer.



```
// area.c
// Function to compute area under  $y = x + 1$  from a to b

double area_under_line(double a, double b) {
    if (a > b) { // swap if needed
        double t = a;
        a = b;
        b = t;
    }
    return 0.5 * (b*b - a*a) + (b - a);
}
```


Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from ctypes import CDLL, c_double

lib = CDLL('./area.so')

# Set argument and return types
lib.area_under_line.argtypes = (c_double, c_double)
lib.area_under_line.restype = c_double

# Example usage
a, b = 2.0, 3.0
area = lib.area_under_line(a, b)

print(fArea under  $y = x + 1$  from {a} to {b} is {area:.6f})
```

Python Code

```
# Define the function
def f(x):
    return x + 1

# Range for the curve
x_vals = np.linspace(1.5, 3.5, 200)
y_vals = f(x_vals)

# Intersection points
points = {
    A: (2, f(2)), # (2,3)
    B: (3, f(3)), # (3,4)
    C: (3, 0), # (3,0)
    D: (2, 0) # (2,0)
}
```

```
# Plot the line  $y = x+1$ 
plt.plot(x_vals, y_vals, label=r"$y=x+1$", color=blue)

# Plot vertical lines  $x=2$  and  $x=3$ 
plt.axvline(2, color=black, linestyle=--, label=r"$x=2$")
plt.axvline(3, color=black, linestyle=--, label=r"$x=3$")

# Plot x-axis
plt.axhline(0, color=black)
```

```
# Shade the region (quadrilateral A-B-C-D)
x_poly = [points[A][0], points[B][0], points[C][0], points[D][0]]
y_poly = [points[A][1], points[B][1], points[C][1], points[D][1]]
plt.fill(x_poly, y_poly, color=skyblue, alpha=0.5, label=Shaded
        Region)

# Mark the intersection points
for name, (x, y) in points.items():
    plt.scatter(x, y, color=red)
    plt.text(x+0.05, y+0.1, f{name}({x},{y}))
```

```
# Labels and legend
plt.xlabel(x-axis)
plt.ylabel(y-axis)
plt.title(Region bounded by  $y=x+1$ ,  $x=2$ ,  $x=3$  and x-axis)
plt.legend()
plt.grid(True)
plt.axis(equal)

# Save the figure
plt.savefig(fig_8.png, dpi=300, bbox_inches=tight)

# Show the plot
plt.show()
```